East West University

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Project Report Task -1

Related Work

Submitted By-

Name	ID
Samia Hasan	2023-3-60-556
Shuvro Dev Sil	2023-3-60-224
Annur Tasnia	2023-3-60-429
Samia Naba	2023-3-60-578

Submitted To-

Dr.Raihun Ul Islam

Associate Professor

Department of Computer Science & Engineering

Literature Review: EMG-Based Movement and Gesture Recognition

Shared column	Paper 1 2024 Human Hand Movement Classification Based on EMG Signal Using Different Feature Extractor	Paper 2 2024 (preprint) EMG-Based Hand Gesture Recognition through Diverse Domain Feature Enhancement and Machine Learning-Based Approach	Paper 3 2022 User-Independent Hand Gesture Recognition Classification Models Using Sensor Fusion	Paper 4 2024 EMGBench: Benchmarking Out-of-Distribution Generalization and Adaptation for Electromyography	Paper 5 2022 Deep Learning and Session-Specific Rapid Recalibration for Dynamic Hand Gesture Recognition from EMG	Paper 6 2025 Electromyographic Typing Gesture Classification Dataset for Neurotechnological Human-Machine Interfaces
Title	Human Hand Movement Classification Based on EMG Signal Using Different Feature Extractor	EMG-Based Hand Gesture Recognition through Diverse Domain Feature Enhancement and Machine Learning-Based Approach	User-Independent Hand Gesture Recognition Classification Models Using Sensor Fusion	EMGBench: Benchmarking Out-of-Distribution Generalization and Adaptation for Electromyography	Deep Learning and Session-Specific Rapid Recalibration for Dynamic Hand Gesture Recognition from EMG	Electromyographic Typing Gesture Classification Dataset for Neurotechnological Human-Machine Interfaces
Authors	Swati Shilaskar, Shripad Bhatlawande, Ranveer Chavare, Aditya Ingale, Rushikesh Joshi, Aditya Vaishale	Abu Saleh Musa Miah, Najmul Hassan, Md. Maniruzzaman, Nobuyoshi Asai, Jungpil Shin	Jose Guillermo Colli Alfaro, Ana Luisa Trejos	Jehan Yang, Maxwell Soh, Vivianna Lieu, Douglas J Weber, Zackory Erickson	Maxim Karrenbach, Pornthep Preechayasomboon, Peter Sauer, David Boe, Eric Rombokas	Jonathan Eby,Moshe Beutel,David Koivisto,Idan Achituve,, Ethan Fetaya,José Zariffa
Year	2024	2024 (preprint)	2022	2024	2022	2025
Dataset	UCI EMG dataset; 8 channels, 3 selected classes, 15,000 samples per class	EMG-based hand gesture dataset,8 channels,3 sec duration,3 sec interval,72 recordings	Myo Armband recordings from 22 participants; 7 gestures and 4 arm positions	Nine curated EMG datasets and a new wearable high-density EMG dataset	Multi-session EMG data and a NinaPro DB5 subset	16-channel bilateral sEMG from 19 participants across two sessions
Application area	Hand movement classification, prosthetics, rehabilitation	EMG-based hand gesture recognition	User-independent gesture recognition, rehabilitation, human–machine interfaces	EMG benchmarking, cross-subject recognition, domain adaptation	Dynamic gesture recognition, assistive technology, human–computer interaction	Typing interfaces, fine-grained gesture recognition, human–computer interaction

Method/model	Notch and Butterworth filtering; sliding windows; frequency-domain features; DWT; feature fusion; Random Forest, KNN, Decision Tree	Diverse-domain feature enhancement and machine-learning classification	EMG and IMU sensor fusion; Adaptive Least-Squares SVM, bilinear models, multilayer perceptron	Benchmark of machine-learning models and preprocessing approaches for out-of-distribution generalization and adaptation	Deep-learning encoder–decoder with session-specific recalibration	Time-domain feature extraction and classification baselines; within-session, cross-session and cross-subject evaluation
Metrics	81.69% accuracy using frequency-domain features with Random Forest	Accuracy, recall, precision, F1 score, specificity, random forest, SVM, gaussian NB, gradient boost, HG boost, decision tree, KNN.	Adaptive LS-SVM: 83.5% with EMG alone and 84.6% with EMG+IMU; MLP: 64.8% and 73.7%, respectively (as summarized earlier; verify protocol).	Results span multiple datasets and among the datasets analyzed (Table 2), the FlexWear-HD dataset achieved the highest LOSO-CV accuracy, registering an 83.9% test accuracy for 10-class classification.	For 11 gestures: 79.5% with recalibration, 49.6% with single-session training, and 55.2% with cumulative training	Best reported feature-set accuracy: 87.4% ± 2.5%;
Strengths	Compares feature extraction strategies and classical classifiers.	Potentially useful for examining complementary EMG feature representations.	Evaluates generalization across participants and the benefit of combining sensor modalities.	Provides a standardized evaluation framework and emphasizes realistic generalization tests.	Shows that session-specific recalibration can improve recognition under session variability.	Provides multi-session data and explicitly evaluates generalization.
Limitations	Only three movement classes are reported in the earlier summary; results depend on preprocessing and data-splitting protocol.	Dataset-specific generalization and preprint status should be considered.	Fusion requires IMU data; performance varies by model and gesture.	Dataset, sensor and protocol differences remain; benchmark results may not transfer directly to a small dataset.	Requires additional calibration data; results depend on the recording protocol and gesture set.	Typing gestures differ from basic movements; participant pool is relatively small.

Research gap	Test whether combining time- and frequency-domain features improves performance on a consistent benchmark.	Test whether feature enhancement transfers to other datasets and movement sets.	Improve user-independent classification without extra sensors or extensive calibration.	Improve generalization to participants and recording conditions unseen during training.	Maintain performance across sessions while reducing re-calibration requirements.	Improve transfer across people and sessions with less calibration data.
Relation to your work	A potential baseline for comparing preprocessing, feature extraction and classical classifiers on UCIS-EMG Basic Movement.	A candidate feature-engineering approach to compare with time-domain and frequency-domain features.	Supports evaluation on unseen users; sensor fusion is only relevant if additional sensor data are available.	Informs train/test splitting and reproducible evaluation for UCIS-EMG.	Motivates a session-robustness experiment if UCIS-EMG contains multiple sessions.	Useful for designing subject-independent evaluation, but not a direct accuracy benchmark for UCIS-EMG Basic Movement.

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